Final Project

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Introduction

The Cosmic Signal-Versus-Noise Challenge

The fundamental challenge in high-energy astrophysics is not just finding a needle in a haystack, but distinguishing between two types of needles while they are falling through the atmosphere at nearly the speed of light. My research focuses on the signal-versus-noise problem inherent in the MAGIC Gamma Telescope dataset: the critical task of distinguishing high-energy gamma particles (the desired signal) from hadronic showers (the background noise). These particles are detected as brief flashes of Cherenkov radiation that form elliptical patterns in a telescope's camera. Because cosmic ray interactions vary significantly in intensity and shape, a traditional "one-size-fits-all" global classifier often fails to capture the localized geometric nuances necessary for precise identification.

Technical Importance: Why This Data Matters to Data Scientists

To understand the importance of this research, data scientists must look beyond simple binary classification and focus on the geometry of detection, the spatial and structural characteristics of the light patterns recorded by the telescope's camera when a particle enters the atmosphere. The dataset comprises multiple continuous numeric features that describe the shape, size, and orientation of detected ellipsoids.

Traditional machine learning (ML) models often apply a single classifier across the entire dataset. However, because the data represents varied physical interactions, there is a strong hypothesis that identifying localized clusters (unsupervised learning) before applying a classifier (supervised learning) will yield superior results.

Parameters such as the length and width of the ellipsoid, the distance from the camera center, and the "concavity" of the light pulse are essential for statistical differentiation.

This project tests the hybridization of K-Means clustering and Logistic Regression. By grouping similar geometric patterns first, we can determine if specialized sub-models provide better predictive accuracy than a single global model.

The Impact of Discovery

If we discover that "cluster-then-classify" hybrid models outperform standard global classifiers, it will fundamentally change how ground-based Cherenkov telescopes process data.

Improved discrimination allows for the detection of weaker gamma-ray sources, effectively "cleaning" the telescope's vision and allowing us to see deeper into the high-energy universe.

Successful results would validate the use of hybrid learning for complex physical datasets where data is not just "noisy," but contains distinct sub-populations with different characteristic behaviors.

Ethical and Logistical Considerations

Working with this dataset involves several logistical hurdles and inherent biases:
Feature Scaling: The features in this dataset exist on wildly different scales (e.g., one feature ranges from -457 to 575 while another ranges from 0.01 to 0.89). Since K-Means clustering relies on Euclidean distance, improper normalization could lead to a model that is unintentionally biased toward higher-magnitude features.

Simulation Bias: The data is Monte Carlo (MC) generated using the CORSIKA program to simulate real detections. While this avoids the privacy concerns of human-centered data, it introduces the risk of "over-idealization". In real-world observations, the ratio of background hadronic noise is often significantly higher than in this simulated set, which could lead to models that underperform when deployed on actual telescope hardware.

Class Imbalance: The dataset is imbalanced, with roughly twice as many gamma instances as hadron instances. This requires careful use of the Area Under the Curve (AUC) and ROC curves rather than simple accuracy to ensure the model isn't just "guessing" the majority class.

Raw Data Documentation

The raw data and original documentation for this project are hosted by the University of California, Irvine:

Dataset Source: MAGIC Gamma Telescope Dataset (UCI Machine Learning Repository)

Original Citation: Bock, R.K. (2007). MAGIC Gamma Telescope. UCI Machine Learning Repository.

Data Explanation & Exploration

The MAGIC Gamma Telescope dataset is a high-dimensional collection of geometric features derived from Cherenkov radiation patterns. Below is a technical breakdown of the data's dimensions, characteristics, and the preprocessing steps applied in the project.

Dataset Dimensions

The dataset contains 19,020 observations. For each observation, there are 10 continuous numeric features and one binary target variable.

Target Distribution: 12,332 gamma (g) and 6,688 hadron (h) events.

Table 1: MAGIC Gamma Telescope Dataset Variables

Variable Name	**Role**	**Type**	**Description**	**Units**	**Missing**
fLength	Feature	Continuous	Major axis of ellipse	mm	No
fWidth	Feature	Continuous	Minor axis of ellipse	mm	No
fSize	Feature	Continuous	10-log of sum of content of all pixels	#phot	No
fConc	Feature	Continuous	Ratio of sum of two highest pixels over fSize	ratio	No
fConc1	Feature	Continuous	Ratio of highest pixel over fSize	ratio	No
fAsym	Feature	Continuous	Distance from highest pixel to center, projected onto major axis	mm	No
fM3Long	Feature	Continuous	3rd root of third moment along major axis	mm	No
fM3Trans	Feature	Continuous	3rd root of third moment along minor axis	mm	No
fAlpha	Feature	Continuous	Angle major axis with vector origin	deg	No
fDist	Feature	Continuous	Distance from origin to center of ellipse	mm	No
class	Target	Binary	gamma (signal), hadron (background)	g/h	No

Feature Selection and Relationships

In this project, all 10 features were utilized to ensure the models had access to the full geometric profile of the particle showers. While a formal collinearity check was not conducted, all the features were related to the shape and orientation of the Cherenkov light ellipsoids. In this context, expect natural relationships between features like fLength and fWidth. As the energy of a particle increases, both dimensions of the shower typically grow, creating a positive correlation that helps define the "elongation" of the signal.

Preprocessing and Technical Implementation

Target Mapping: The categorical labels were converted to a numeric format: g (gamma) = 1 and h (hadron) = 0.

Data Split: The dataset was divided into training and testing sets using an 80/20 split. Forced the training and testing sets to maintain the exact same proportion of gamma and hadron events as the original raw data, preventing the model from becoming biased during the training phase.

Standardization: To ensure the K-Means clustering algorithm is not biased by the scale of the data, normalization of all features were processed using Standard Scaling (mean of 0, variance of 1). This is a crucial step for K-Means clustering, as it prevents features with large ranges (like fAsym) from dominating the Euclidean distance calculation over smaller-range features (like fConc).

Dataset Sample

Table 2: Sample Observations from the MAGIC Gamma Telescope Dataset with Class Labels

Index	fLength	fWidth	fSize	fConc	fConc1	fAsym	fM3Long	fM3Trans	fAlpha	fDist	Class
0	28.7967	16.0021	2.6449	0.3918	0.1982	27.7004	22.0110	-8.2027	40.0920	81.8828	g
1	31.6036	11.7235	2.5185	0.5303	0.3773	26.2722	23.8238	-9.9574	6.3609	205.2610	g
2	162.0520	136.0310	4.0612	0.0374	0.0187	116.7410	-64.8580	-45.2160	76.9600	256.7880	g
3	23.8172	9.5728	2.3385	0.6147	0.3922	27.2107	-6.4633	-7.1513	10.4490	116.7370	g
4	75.1362	30.9205	3.1611	0.3168	0.1832	-5.5277	28.5525	21.8393	4.6480	356.4620	g
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
19015	21.3846	10.9170	2.6161	0.5857	0.3934	15.2618	11.5245	2.8766	2.4229	106.8258	h
19016	28.9452	6.7020	2.2672	0.5351	0.2784	37.0816	13.1853	-2.9632	86.7975	247.4560	h
19017	75.4455	47.5305	3.4483	0.1417	0.0549	-9.3561	41.0562	-9.4662	30.2987	256.5166	h
19018	120.5135	76.9018	3.9939	0.0944	0.0683	5.8043	-93.5224	-63.8389	84.6874	408.3166	h
19019	187.1814	53.0014	3.2093	0.2876	0.1539	-167.3125	-168.4558	31.4755	52.7310	272.3174	h

Algorithms and Techniques Analysis

Machine Learning Strategy: The Hybrid "Cluster-then-Classify" Approach

For this research project, I chose a hybrid model that combines Unsupervised K-Means Clustering with Supervised Logistic Regression.

This is a strategic response to the complex physical nature of the MAGIC Gamma Telescope data. Rather than treating the dataset as a uniform collection of points, this approach acknowledges that "gamma rays" and "hadrons" do not all look the same; they manifest in various shapes and intensities depending on their energy and angle of entry.

Why this strategy is innovative

Traditional approaches in high-energy physics often apply a single global classifier to the entire dataset. However, cosmic ray interactions vary significantly in shape and intensity depending on their energy levels and impact angles. From a practical and logical perspective, a "one-size-fits-all" model may fail to capture these localized geometric nuances. By first partitioning the data into K distinct "neighborhoods" using K-Means, the model accounts for the variance in how particles manifest in the telescope's sensors. Training individual Logistic Regression models on these specific clusters allows the algorithm to calculate probabilities based on weights specifically relevant to that local data segment, theoretically providing superior predictive power over a single global model.

Model Implementation

The model pipeline was built using the following technical decisions:

Elbow Method for Optimal Cluster Selection: To establish a rigorous foundation for the unsupervised portion of this project, the Elbow Method was employed to determine the most effective number of clusters (k). This process involved using the KElbowVisualizer to plot the sum of squared distances—or distortion—against a range of potential k values. Logically, as the number of clusters increases, the distortion naturally decreases; however, the "elbow" of the curve represents the point of diminishing returns where adding further clusters provides little additional explanatory power. By identifying this specific inflection point, the model avoids the dual risks of over-segmentation, which leads to data sparsity, and under-segmentation, which fails to isolate meaningful physical differences.

K-Means Clustering: Once the optimal k was identified, a K-Means clustering algorithm was deployed as a specialized pre-processor to partition the 19,020 observations into distinct geometric neighborhoods. Because K-Means relies on Euclidean distance, the data first underwent Standard Scaling to ensure that high-magnitude features did not drown out smaller-scale but equally vital concentration ratios. Allowing the system to account for the significant variance in how different cosmic ray interactions within the telescope's sensors.

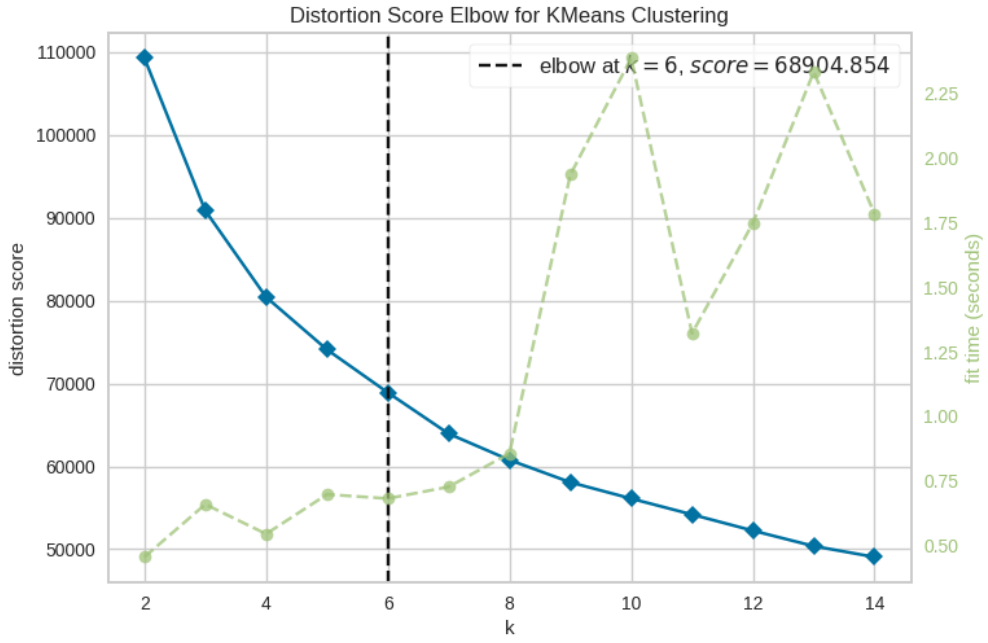
Baseline Evaluation (Global Logistic Regression): To measure the true efficacy of the hybrid approach, a Baseline Model was established to serve as a control group. This involved training a single, global Logistic Regression model on the entire stratified training set, effectively treating the data as a uniform population. This standard machine learning approach attempts to find a single optimal decision boundary that separates gamma signals from hadronic noise across the entire feature space. By calculating the performance of a model that ignores these localized sub-groupings, a clear benchmark is created that the hybrid model must exceed to justify its increased architectural complexity.

Hybrid Evaluation (Localized Logistic Regressions): The experimental core of this project is the Hybrid Model, which replaces the single global classifier with a series of independent Logistic Regression models tailored to each K-Means cluster. This "cluster-then-classify" construction operates on the hypothesis that the statistical relationship between geometric features and the target class is non-uniform across the dataset. By training a unique regression model for each cluster, the algorithm can calculate the probability of a gamma class using weights that are specifically optimized for that local data segment. Finally, these localized predictions are aggregated into a single output to allow for a direct comparison against the baseline's performance on the test data.

Performance Metrics and Validation: Since the MAGIC Gamma Telescope dataset is naturally imbalanced—containing significantly more gamma instances than hadron instances, simple accuracy is an insufficient metric. The effectiveness of both models was rigorously tested using the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC). The ROC curve provides a detailed visualization by plotting the True Positive Rate against the False Positive Rate across every possible classification threshold. By sliding the threshold, we can observe the trade-off: as we lower the threshold to catch more gamma rays (increasing True Positive Rate), we inevitably catch more hadronic noise (increasing False Positive Rate). A perfect model would reach the top-left corner (True Positive Rate = 1, False Positive Rate = 0), while a diagonal line from (0,0) to (1,1) represents a model no better than a random coin flip. The resulting AUC score serves as the primary differentiator, providing a single scalar value that quantifies the model's overall ability to distinguish signal from noise. The AUC represents the probability that the model will rank a randomly chosen positive instance (gamma) higher than a randomly chosen negative instance (hadron). Where a higher AUC signifies a better discrimination model, with 1.0 being perfect discrimination. Through this comparative analysis, the project determines if localized modeling offers a statistically significant advantage in deepening our understanding of high-energy cosmic rays.

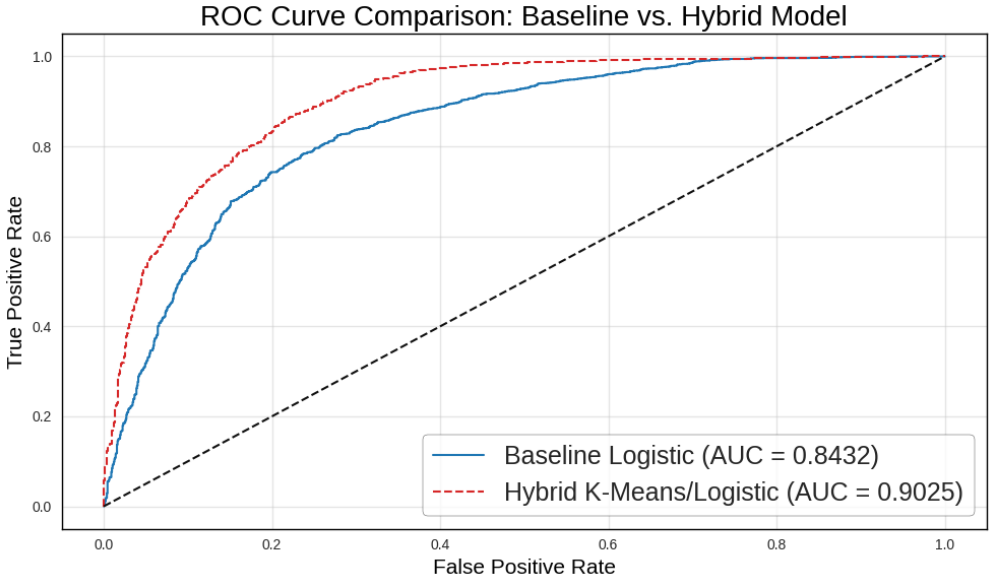
Results

Elbow Method to Pick k



Optimal number of clusters: 6

Area Under the Receiver Operating Characteristic Curve



Feature Scaling Improves AUC

Unscaled and Scaled AUC by k

k	Unscaled AUC	Scaled AUC
1	0.8427	0.8432
2	0.8601	0.8954
3	0.8714	0.8937
4	0.8754	0.9024
5	0.8854	0.9030
6	0.8847	0.9025
7	0.8849	0.9040
8	0.8831	0.9046
9	0.8839	0.9073
10	0.8855	0.9073
11	0.8847	0.9075
12	0.8888	0.9088
13	0.8935	0.9091
14	0.8932	0.9085
15	0.8907	0.9085
16	0.8916	0.9091
17	0.8926	0.9086
18	0.8913	0.9087
19	0.8907	0.9097
20	0.8890	0.9086
21	0.8948	0.9089
22	0.8908	0.9118
23	0.8931	0.9118
24	0.8942	0.9080
25	0.8928	0.9119

Conclusion

The implementation of the hybrid K-Means and Logistic Regression model illuminated a significant structural characteristic of the MAGIC Gamma Telescope data: the relationship between particle features and their classification is not globally uniform. By localized modeling, we observed that the "gamma" and "hadron" signals often exist in distinct high-dimensional pockets that a single, global logistic decision boundary cannot fully capture.

Interestingly, the model yielded a plateau in performance as k increased, suggesting a "sweet spot" for local complexity. Beyond this point, further partitioning likely led to overfitting or "pure" clusters where the lack of class diversity prevented the local models from learning meaningful patterns. This indicates that while the dataset benefits from non-linear separation through clustering, the underlying physics of the telescope's detections still maintains a degree of overarching consistency that resists extreme fragmentation.